Industrial Energy-Related Technologies and Systems iets

An IEA Technology Collaboration Programme

Task XXIV – Subtask 1

Participants meeting

May 23rd 2025

Where are we...

1. Kick-off meeting **14th March 2025**

- ❖ Introduction of the Task
- ❖ Outline the work packages (WPs),
- ❖ Discussion/rules of the different ways of contributing to this Subtask.

Recording/Slides of the kickoff available

2. Partners Presentations meeting **11th April 2025**.
During this meeting, each partner presented:

- ❖ Introduction to their activities related to the industry decarbonization.
- ❖ Pre-selection of the activities in which are willing to contribute.
- ❖ Candidacy of participants as activity coordinators.
- ❖ Miro Board

Add your name in the Participants field next to the Activity description.

Consider mentioning your interest in being an Activity coordinator.

3. Executive Committee meeting **15th May 2025**: Task XXIV development updates.

4. Participants meeting **23rd May 2025**: Discussion about Participation, Sharing, Dissemination.

- Industrial sectors and industrial process models
- Decarbonization technologies
- Process integration models
- Energy systems models
- Life cycle sustainability assessment models
- Roadmaps generation and dissemination

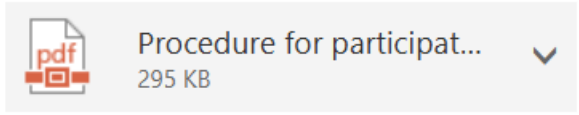
Agenda

1. Instructions about official as participants registration (template and info).
2. Introduction to model sharing, ontologies and procedure (Activity 1).
3. Models inventory & models database (Activities 2 and 3).
4. ESCAPE 2025 Semi-plenary invitation.
5. Conferences (EFCE, ECOS, ESCAPE, ECCE, EUBCE).
6. Workshop December Portugal (Paul Stuart's invitation).
7. Varia

Participants registration (template and info).

Participant organizations (POs) → Task managers (TMs) → IETS Secretariat (IS) → Contracting Party (CO) → TMs & POs





https://iea-industry.org/members/

Participation of groups in Switzerland

Date of notification	2025-05-23
First notification for Task/subtask or update?	☒ First notification ☐ Update
IETS Task number:	XXIV
Subtasks number(s):	1
Responsible Contracting Party (CP) (i.e., member country or sponsor):	Switzerland
Task Manager:	Daniel Florez-Orrego, François Maréchal

Letter of notification of participation in IETS Task/Subtask

Confirmed Participating Organisations:	**Lead contact:**
Consortium consisting of: EPFL Switzerland (task manager) HES-SO Wallis Valais (engineering school) NetZero Lab Sierre (official consortium) Swiss Federal Office of Energy (gov. partner) PinchSmart (subcontractor to official consortium) MAS SESE (as an associated partner)	Daniel Florez-Orrego, François Maréchal École polytechnique fédérale de Lausanne daniel.florezorrego@epfl.ch francois.marechal@epfl.ch

Means of contribution by type and duration *(Important for Task Manager to plan activities)*
e.g. self-financed; self-financed via dissemination channel of a research project; consortium pending; consortium paid by national funding; as long as background project runs, until national funding period expires, etc
EPFL, HES-SO and NZL are financed by Novelis Inc, Oiken, and the Swiss Federal Office of Energy. MAS SESE is an associated partner which does not receive funding but is treated as a full partner unless revocation from either side (i.e. consortium or partner). The funding expires on 2027-11-01. Limited self-financed contribution is possible afterwards.

Task structure & term

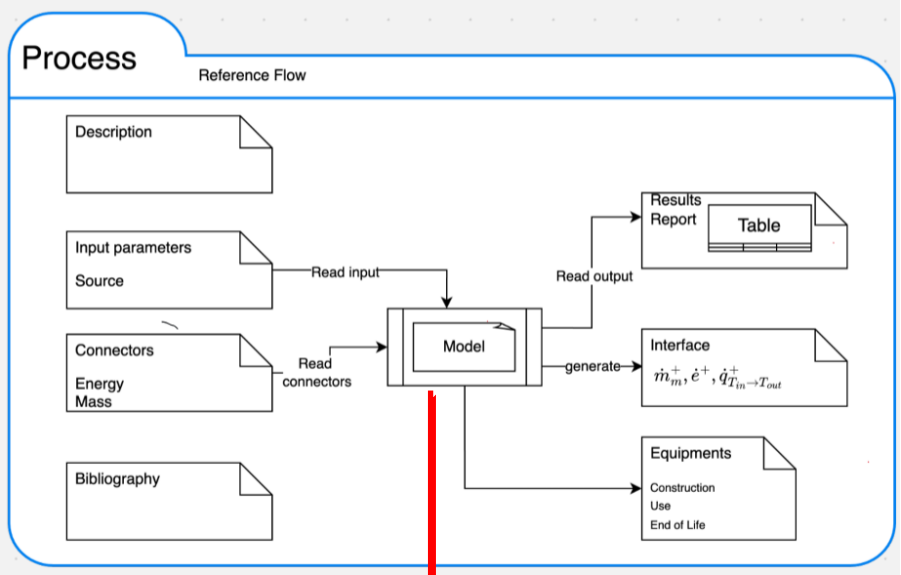
Current time frame: 01.01.2025 – 01.11.2027

Task Managers: Daniel Florez-Orrego, François Marechal, EPFL, Switzerland

Sub-task	Content	Sub-task Managers
1	*Enabling a Shared Database of Ex-Ante Models*: Model-database sharing framework. Demonstrate the role of process integration for decarbonizing the industrial sectors. Critical aspects, such as housing, maintenance and curation of the database are discussed.	Daniel Florez-Orrego, François Maréchal, EPFL, Switzerland Collaborators: Oktay Boztas, Meire Ribeiro Domingos, Pullah Bhatnagar, Cecilia Mannik, Vibhu Baibhav, Du Wen

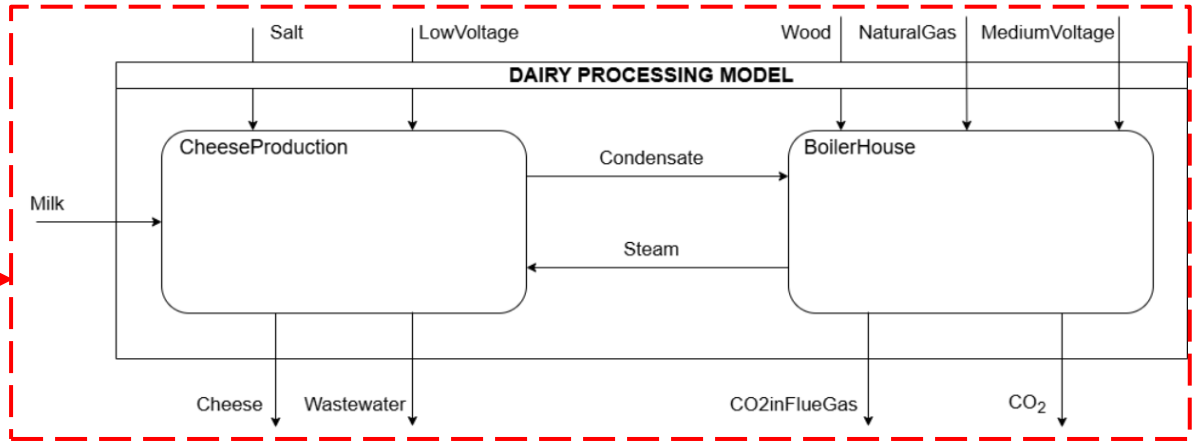
Model sharing, ontologies and procedures (Activity 1).

Excel template proposal



Model Name	DairyProcessing
Authors and Contributors	
Version	1.0
Confidentiality	Open
Creation Date	
References	
Grade	GRAY-BOX
Software	DWSIM
URL	
Last Update Date	
Keywords	Food
Reference Capacity (main product/feedstock flowrate)	10 kg/h Cheese

Metadata for univocally defining the version, authorship, confidentiality, grade of detail, reference capacity and software used.



Model sharing, ontologies and procedures (Activity 1)

EXTERNAL CONNECTORS							
Unique Identifier Number, UID	NAME (ALIAS)	TYPE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK-FAT3	MILK	MASS	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
CHEESE-SOFT	CHEESE	MASS	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
SALT-NACL	SALT	MASS	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
WASTEWATER-SEWAGE1	WASTEWATER	MASS	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
TAP-WATER	WATER	MASS	kg/h	25	1	MASSFRAC:H2O[1]	-1 (L)
WOOD-DRY	WOOD	MASS	kg/h	25	1	MASSFRAC:C[0.47]&H[0.06]&O[0.45]&S[0]&ASH[0.02]	-1 (S)
NATURALGAS	CH4	MASS	kg/h	5	1	MASSFRAC:CH4[1]	-1 (V)
ELEC-50KV	MEDIUMVOLTAGE	ELECTRICITY	kW	-	-	-	-
ELEC-13KV	LOWVOLTAGE	ELECTRICITY	kW	-	-	-	-
FLUEGAS-M	CO2INFLUEGAS	EMISSIONS	kg/h	120	1	MOLEFRAC:CO2[0.15]&H2O[0.12]&N2[0.73]	-1 (V)
CO2-PURE	CO2	EMISSIONS	kg/h	25	1	MASSFRAC:CO2[1]	-1 (V)

Interfacing the specific models with the global database of connectors

Engineering variables must be physically defined, determined (software, formula, etc.), represented and referenced

VARIABLES												
Name	Type	Full name	User grade	Default Value / Formula	Physical unit	Min bound	Max bound	Multiperiod	Data Source	Latex syntax	Bibliography	Comment
Tmilk	DECISION	Milk inlet temperature	Mandatory	4	C	3	20	P,T,O(S)	./milkplant_data.csv	T_{milk}	@Florez-Orrego et al	Infeed temperature of the milk from transportation truck
Ta	CONTEXT	Environmental temperature	Optional	25	C	-273 C	1000	P,T,O(S)	./milkplant_data.csv	T_{a}	@Boztas et al	Environmental temperature used for the calculation of the PV efficiency (it is overwritten by the GIS information)
R	LITERATURE	Constant of the ideal gases	Private	8.314	kJ/kmolK	-	-	P,T,O(S)	./milkplant_data.csv	R_{gas}	@Ribeiro Domingos et al	Contant of the ideal gases
Lactose	MEASURED	Lactose fraction	Optional	2%	-	0%	5%	P,T,O(S)	./milkplant_data.csv	-	@Bhatnagar and Baibhav	Content of Lactose of the milk

Model sharing, ontologies and procedures (Activity 1)

unit name	Minimum Capacity Multiplier	Maximum Capacity Multiplier
CheeseProduction	1	1

Reference sizes and upper/lower caps are important to establish the implementation range of the model.

CONNECTORS								
NAME (ALIAS)	TYPE	DIRECTION	VALUE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK	MASS	IN	100	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
SALT	MASS	IN	1	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
CHEESE	MASS	OUT	10	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
WASTEWATER	MASS	OUT	91	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
LOWVOLTAGE	ELECTRICITY	IN	3	kW	-	-	-	-
STEAM	MASS	IN	50	kg/h	SAT	3	MASSFRAC:H2O[1]	1 (V+L)
CONDENSATE	MASS	OUT	50	kg/h	SAT	3	MASSFRAC:H2O[1]	0 (V+L)

HEAT STREAMS								
name	Type	Tin [C]	Tout [C]	Hin [kW]	Hout [kW]	dTmin Contribution [C]	alpha [kW/m2K]	HeatCascade
pasteurization_heating	Cold	25.0	70.0	0.0	5.0	5	0.50	CUSTOM
pasteurization_quenching	Hot	70.0	25.0	5.0	0.0	5	0.50	CUSTOM
evaporation_preheating	Cold	25.0	90.0	0.0	2.0	5	0.50	DEFAULT
evaporation_latentheat	Cold	90.0	90.0	0.0	18	2.5	1.00	DEFAULT
condensation_latent	Hot	90.0	90.0	18.0	0	2.5	1.00	DEFAULT
condensation_subcooling	Hot	90.0	25.0	2.0	0.0	5	0.50	DEFAULT
cheese_curation	Cold	25.0	30.0	0.0	5.0	8	0.10	DEFAULT

Pinch analysis (a.k.a. heat integration) requires the knowledge of the heat cascade and can include different heat cascades (constraints).

EQUIPMENTS				
name	Type	Parameters	Value	units
pump_milk	Pump	Power	2	kW
		Lifetime	20	years
		Date	2025	year
HEX_101	Heat Exchanger	Area	3000	m2
		Pressure	1	bar
		Lifetime	20	years
		Date	2025	year
Stirrer	Asset	Capital	75000	EUR
		Lifetime	20	years
		Date	2025	year

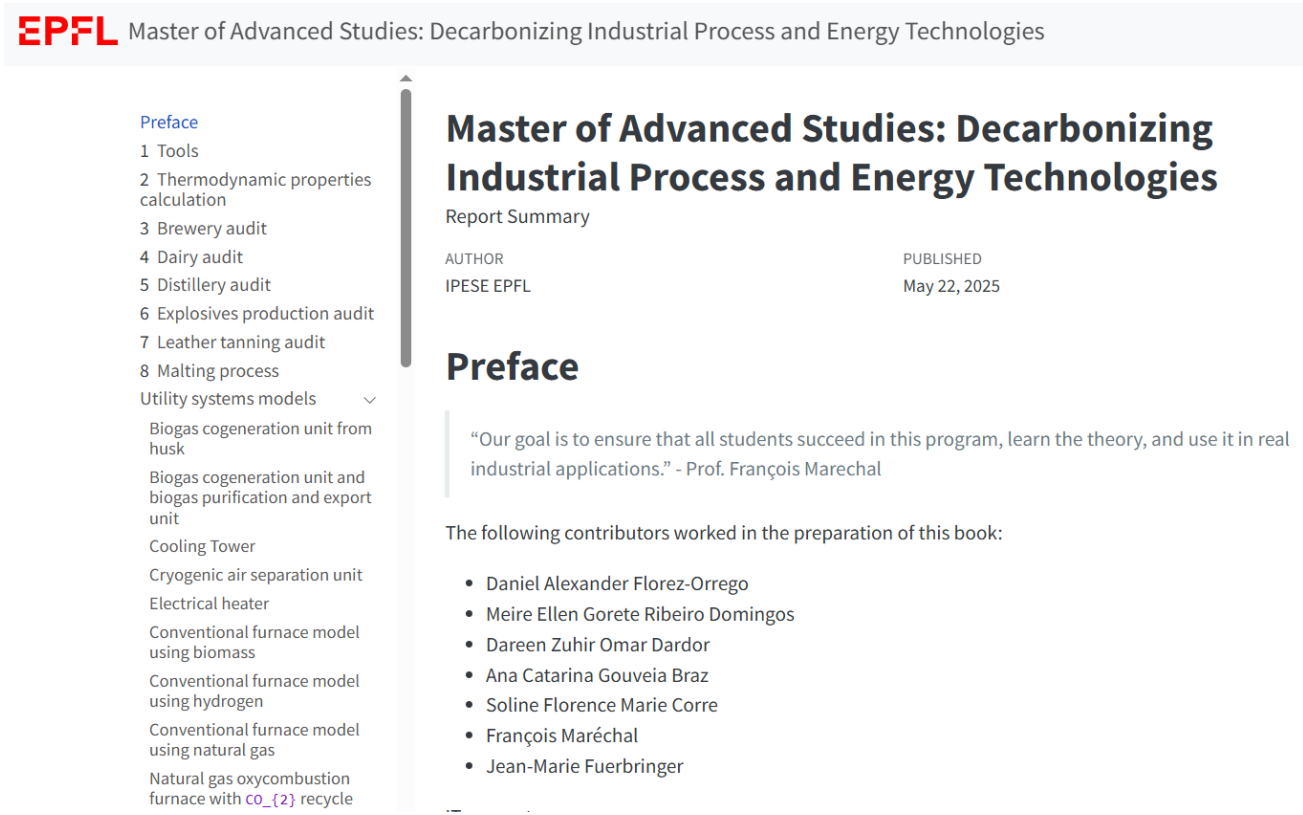
Reference sizes are important to adequately calculated the costs and account for economies of scale.

Models inventory & models database (Activities 2 and 3)

EPFL models database

- MASbook database
- EPOS database
- AIDRES database

Data + Model
Storage
Versioning
Curation
Intellectual property
Confidentiality
Software



Participants interests

Name	Uni	Country	Procedures	Processes	Technologies	Energy System	Process integration	LCA metrics	Roadmaps	Dissemination
Henri Matos (IST - PT)	IST	PT	Ontologies and Standardization	CO2toMethanol, CO2toDMC	CCU		Aspen Plus	Processes LCA		
Fernando Martins (Porto - PT)	Porto	PT		Gasification, Refinery, Microalgae, Biological			Energy Efficiency, Machine Learning,		Strategy Improvement	
Gregoire Leonard (ULg - BE) Muhammad Salman	ULG	BE		Sugar, glass, lime, cement, steel	Decarbonization, CCUS		Validation, Open-access flowsheeting, OSMOSE Platform	Societal impact of energy transition		Documentation, Visualisation(indicate.com)
Mar Perez Fortes (TuDelf - NL)	TuDelft	NL	Excel files, aspen models, cost data	Petrochemicals, plastics			Aspen Plus, Pyomo, Gurobi, MILP			
Simon Harvey (Chalmers - SE) Judit Fortet Casabella	Chalmers	SE		Chemicals, Refinery, Gasification, Methane	CCUS, Electrification		Experimental validation of gasification and gas cleaning		Transition Pathways	
Henrik Thunman	Chalmers	SE								
Assad Zoughaib (MinesParisTech- F)	MinesParisTech	FR		Agrofood, Pulp and Paper	HP, MVR, CCS(amine)		Aspen Plus, Modelica, Python			
Pablo Silva (TUM - DE)	TUM	DE		Thermochemical conversion, SAF, Ethanol			Aspen Plus			
Flavio Manenti (Polimi - IT) Mattia Vallerio	Polimi	IT		Gasification, Chemicals, Methanol			Aspen Plus, Pinch Analysis			
Rene Hoffmann (TuWien - AT)	TuWien	AT					Industrial Excellence Heat Recovery (Task 15)			
Benjamin Zühlsdorf (Teknologisk - DK)	Technologisk	DK			HP					

Participants interests

Nasrin Kermani (DTU - DK)	DTU	DK								
Roger Padulles (DTU - DK)	DTU	DK		Petrochemical, Food and Beverage			Python, Modelica			
Zheng Grace Ma (SDU - DK)	SDU	DK					Incorporate advanced digital and AI-based technologies		Resiliance and adaptation under uncertainty, TASK 18, 21, 22	
Thomas Adams (NTNU - NO)	NTNU	NO	Standards	Chemicals, Biorefinery	HP, CCU		Aspen Plus, COIN-or			PSE Community.org https://psecommunity.org/materials/calculationfiles
Sara Russo-Garrido (CIRAIG - CN)	CIRAIG	CN						Processes LCA	Roadmaps and Metrics	
Daniel Florez-Orrego (EPFL) Oktay Boztas Meire Domingos Vibhu Baibhav Pullah Bhatnagar Wen Du	EPFL	CH	Database, Sharing, Anonymization	Fertilizers, SAF, aluminium, plastics, AIDRES	Gasification, co-generation, Power2X, SOCs, CCUS	x	Aspen Plus, Pinch Analysis, OSMOSE Platform	x	x	x

Add your name in the Participants field next to the Activity description.

Consider mentioning your interest in being an Activity coordinator

ESCAPE 2025 Semi-plenary (dissemination)

Invitation to the ESCAPE 2025 Semi-plenary presentation.



Systems & Control Transactions

Research Article - Peer Reviewed Conference Proceeding
ESCAPE 35 - European Symposium on Computer Aided Process Engineering
Ghent, Belgium. 6-9 July 2025
Jan F.M. Van Impe, Grégoire Léonard, Satyajeet S. Bhonsale, Monika E. Polańska, Filip Logist (Eds.)

PSE PRESS

Process integration for industry decarbonization: Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

Daniel Flórez-Orrego[1], Muhammad Salman[2], Oktay Boztas[3], Dareen Dardor[4], Meire Ribeiro-Domingos[5], Pullah Bhatnagar [6], Gregoire Leonard[7], Francois Marechal[8]

[1,3,4,5,6,8] Industrial Process and Energy Systems Engineering, EPFL, Sion, 1950, Switzerland
[4] Institute of Sustainable Energy, University of Applied Sciences and Arts Western (HES-SO), Sion, Switzerland
[2,7] Department of Chemical Engineering, University of Liège, Belgium
*Corresponding Author: daniel.florezorrego@epfl.ch

ABSTRACT

The mass adoption of process simulators has accelerated data generation for the

Conferences (EFCE, ECOS, ESCAPE, ECCE, EUBCE).

EFCE



ECOS



ESCAPE



ECCE



EUBCE



Workshop December Portugal (Paul Stuart's invitation).

❖ Invitation Paul Stuart:

- Meeting in Lisbon for a Working Meeting before the conference, Monday 1 December and/or Tuesday 2 December, this opportunity will be possible. The IBBC25 Conference Chair, Professor Jorge Costa, can arrange meeting rooms and catering for a reasonable fee at his university - Instituto Superior de Educação e Ciências (ISEC) – where the conference will be held.

- Technical Session as Part of the Industry-Based Biorefineries for Sustainability IBBC2025 in Lisbon, 3rd – 5th December, 2025.

❖ First working Meeting Valencia 13th June, 2025, EUBCE Conf. → Collaboration with Task XI

Dissemination

Participation in the online TCP- ETSAP & IETS Workshop on April 10, 2025



- Mailing list updated (...continuously).
- Whitelisting EPFL Task managers emails (so that you can receive the news ☺)
- Instant messaging tool (Slack, Mattermost, etc.)
- Google Form (Questions to keep the momentum)

Questions!

- How many models (process and technologies) will you use as a starting point?
- How much time translating those models into the suggested format would be necessary?
- Are the template suggestions compatible with the modeling paradigms/structures in your labs?
- In negative case, what would it need to be added or removed to make it compatible?
- Does the proposed template miss any information or have irrelevant additional one?
- Does it need a clearer distribution in terms of tabs, fields and data input format?
- Should alternative templates be proposed per groups of 2 member countries to be analyzed in plenum?
- Does the suggested template have a bias/constraint to address different types of models or solving methods (linear, non-linear, metaheuristic, black-grey-white boxing, etc.)?
- How much is the confidentiality an issue to share models?
- How does the intellectual property play a role in openly (or bilaterally) share robust models (published or unpublished)?
- Is scientific citation enough for leveraging the model? Should there be in place a cooperative publication strategy in which the original authors of the models must be added as co-authors?
- Are there legal regulations in a member country that limit (or endorse) sharing intellectual property data between researchers, industries or policy makers?
- Do the restrictions associated to sharing proprietary-software models files have so far prevented sharing models with international collaborators?
- Which is the preferred models inventory methods/storage tools currently used by your institution/group (considering a combination of software files, database formats, templates, etc.)?
- Which is your expectation about contributing and accessing a large database of shared model:
- Consult validated models and use them as they are.
- Compare with your own models and suggest improvements.
- Share with you faculty members (students, colleagues) for educational purposes.
- Use the models in industrial-related projects
- Impart short courses (on decarbonization and electrification)
- Be acknowledged as a publisher of data, data in brief, modelling data
- Altruism.
- Which is the best procedure to peer-validate the database of models?